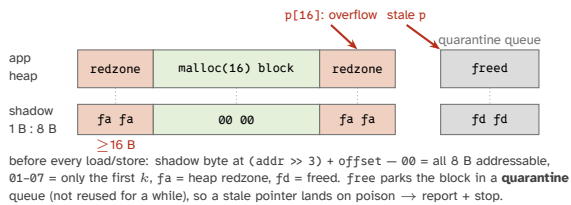


In one line: Memory and threading bugs corrupt silently and crash *later, elsewhere*. Sanitizers are **compiler instrumentation + a runtime** that check every access as it happens and report the **first bad access** with the stacks that explain it. Runtime checkers — Main Thread Checker (MTC), Thread Performance Checker (TPC), exclusivity — need no special build. ASan and TSan cost 2–20× and their runtimes are not meant for production: Debug schemes, test plans, CI. Hardware memory tagging is the one built to ship.

ASan: shadow memory + redzones + quarantine



TSan: is there a happens-before edge?



per 8 B of memory, **4 shadow slots** keep recent accesses (thread, epoch, read/write); a full cell overwrites a random slot. Every thread and every lock / queue / atomic carries a **vector clock**: release -> acquire is an edge. Same address, ≥ 1 write, no edge -> race - even if this run *happened* to be fine.

Tool → catches → cost → where

tool	catches	cost	where
ASan	heap / stack / global overflow, use-after-free, use-after-scope, double + invalid free, use-after-return opt-in. C + Swift	2–5× CPU, 2–3× mem	device + Simulator
TSan	data races, Swift access races, NS/CF mutable-collection races, bad mutexes, thread leaks	2–20× CPU, 5–10× mem	Simulator, 64-bit macOS
UBSan	C UB: divide by 0, misaligned, null, signed overflow, shift, bool / enum, bounds. Not Swift (Swift traps itself)	≈20% CPU	with ASan or TSan
MSan	uninitialised reads	3×	not Apple
MTC	UIKit / AppKit / WebKit API off main	1–2%; launch + ≤100 ms	on by default
TPC	priority inversion; sync I/O + networking on main	minimal	Run; macOS + iOS
HW tags	use-after-free, out-of-bounds: pointer tag ≠ memory tag	ships	A19+, M5+ chips
Exclusivity	overlapping inout / mutating access (Swift)	small	Release too
Dynamic isolation	actor-isolated code called off its actor	crash, not race	Swift 6 mode

Malloc debugging (scheme checkboxes): *Scribble* alloc -> 0xAA, free -> 0x55 · *Guard Edges* a guard page each side of blocks ≥ 4096 B · *Guard Malloc* own VM page per alloc, freed page removed (Mac + Simulator) · *Malloc Stack* alloc backtraces in the Memory Graph.

How they work

- ASan** replaces malloc/free, checks both buffers of memcpy-style calls; halt_on_error=1: the first report ends the run. Misses leaks, uninitialised reads, integer overflow. -O1 trims its memory (TSan too).
- TSan** instruments accesses *and* sync calls; halt_on_error=0: it keeps running, one report per race. Swift 6 mode rejects races at compile time, but @unchecked Sendable and nonisolated(unsafe) switch that check off — TSan still sees them, and C / ObjC.
- MTC / TPC** swap system methods for thread-checking variants — no recompile; libMainThreadChecker.dylib can be injected into an existing macOS binary. TPC: dispatch_semaphore_wait / dispatch_group_wait cannot donate priority -> inversion; silence known ones with PERFC_SUPPRESSION_FILE.
- Zombies**: freed objects stay as zombies (NSDeallocateZombies=NO): “message sent to deallocated instance” + trap. Field signs: objc_msgSend / objc_retain / objc_release on top, or an unrecognised selector (memory reused).
- HW tags**: Enhanced Security -> *Enable Hardware Memory Tagging* (...checked-allocations entitlement); *soft mode* writes a simulated crash report instead of killing — off before users get the build.
- Exclusivity**: static where provable, dynamic (swift_beginAccess) for class properties, statics, globals, escaping closures; -enforce-exclusivity-unchecked drops the Release checks.
Dynamic isolation (SE-0423): @objc thunks, @preconcurrency conformances, closures passed to unaudited APIs; -disable-dynamic-actor-isolation opts out.

Since

Xcode 7 ASan · **8** TSan · **9** UBSan, MTC, Swift access races · **Swift 5** exclusivity checks in Release · **Xcode 14** TPC · **Swift 6** dynamic actor isolation · **Xcode 26** runtime issues in tests, no Zombies template, C++ container-overflow on · **Xcode 27** ASan on 27.0 OSes needs ≥ 26.5 .

Turning them on

clang -fsanitize=address (or thread, undefined) · swiftc -sanitize=address (or thread; no UBSan) · xcodebuild test -enableAddressSanitizer YES (...Thread..., ...UndefinedBehavior...). Scheme -> Run -> **Diagnostics**; tests: test-plan configurations, one per sanitizer. *Runtime API Checking* warns by default; *On (as Failure)* fails the test. UBSan continues after a report; -fno-sanitize-recover exits, -fsanitize-trap traps (SIGILL / SIGTRAP, no runtime). Container-overflow noise: ASAN_OPTIONS=detect_container_overflow=0 in the Run scheme and each ASan test plan — build settings cannot switch it off.

Reading a report (ASan, abridged)

```
==4242==ERROR: AddressSanitizer: heap-use-after-free
READ of size 8 at 0x6020000eff0 thread T0
#0 parse_header buf.c:42 // 1 WHERE: bad access
0x60200000eff0 is located 0 bytes inside of 16-byte region
freed by thread T0 here:
#1 conn_close conn.c:88 // 2 WHO freed it
previously allocated by thread T0 here:
#1 conn_open conn.c:31 // 3 WHO made it
SUMMARY: AddressSanitizer: heap-use-after-free buf.c:42
```

Bug type first, then three stacks: access, free, allocation — the fix is usually in #2 or #3, not #1. More shadow bytes: f1/f2/f3 stack left / mid / right redzone · f5 stack after return · f8 after scope · f9 global redzone · fc container overflow.

TSan WARNING: ThreadSanitizer: data race: *Write ... by thread T3 + Previous read ... by main thread* — two stacks, two threads, one address, no edge: fix the *ownership* (actor, serial queue, Mutex), not the timing. Other kinds: Swift access race · thread leak · lock-order-inversion (potential deadlock) · invalid or unlocked mutex · signal-unsafe call inside of a signal.

Exclusivity trap at run time

```
final class Counter { var n = 0 }
func bump(_ x: inout Int, by c: Counter) { x += c.n }
let c = Counter()
bump(&c.n, by: c) // modify of c.n open, then a read of c.n
// Simultaneous accesses to 0x..., but modification requires
// exclusive access. Previous access (a modification) started
// at ... Current access (a read) started at: swift_beginAccess
```

Traps

- ! ASan + TSan in one run — impossible: two test-plan configurations, two CI runs.
- ! Clean TSan run ≠ race-free: unexecuted interleavings and overwritten slots go unseen.
- ! Crash on 0x400000000000bad0: synthesized nonatomic ObjC setters briefly store this sentinel — two threads mutated the property.

Remember

ASan = where memory, TSan = when threads, UBSan = C rules, MTC / TPC = the main thread, tags = the one that ships. Crash here, not later.